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CodioProjectFileEditFindViewToolsEducationHelpConfigureProject Index (static)Configure

box.txttemp.pyx1# WRITE YOUR CODE HERE2import secret3import os4import openai56openai.api_key=secret.api_key7prompts = ''8response = openai.Completion.create(model="text-davinci-002", prompt=prompts)9print(response['choices'][0]['text']).strip()10

Codio - Introduction to GPT-3

SANDIPAN DEY

Guide

Collapse

Models

To query the OpenAI API, we call the `Completion.create` function and store its return value in the variable `response`.

```
response = openai.Completion.create(model="text-davinci-002", prompt=prompts)
```

We can see the `Completion.create` function takes a `model` keyword argument and a `prompt` keyword argument. The `prompt` keyword argument is the variable which we use to store the prompt we are going to pass to the GPT-3 model.

Now let's tackle the `model` keyword argument. When OpenAI defines GPT-3, it defines it as a set of models that can understand and generate natural language. GPT-3 offers four main models suitable for different tasks. For example, the `ada` model is the fastest, whereas the `davinci` model we used earlier is the most capable.

Model	Description	Max Request	Training Data
text-davinci-002	Most capable GPT-3 model. Can do the task of the other models, often with less context. Contains the most recent training data. Additionally, this model is capable of inserting completions within a text.	4000 tokens	up to June 2021
text-curie-001	Faster and lower cost than the <code>davinci</code> model, but it is less capable.	2048 tokens	Up to Oct 2019
text-babbage-001	Works best with straightforward tasks only. It is fast and low cost when compared to the previous two models.	2048 tokens	Up to Oct 2019
text-ada-001	Lowest cost and fastest model. You get the best results with <code>ada</code> when the tasks are straightforward.	2048 tokens	Up to Oct 2019

OpenAI plans to continuously improve their models over time. `davinci` is generally the most capable model. `curie` can perform many of the same tasks as `davinci` but faster with a reduced cost. As we continue to experiment with GPT-3, we will use the `davinci` model since it yields the best results. Once things are working, we can try different models to see how the results differ.

Add the following code to the IDE. Make sure to delete the previous `prompt`, `response` and `print` statements. We should only have our libraries and our key left. We are going to have different variables representing the responses from each of the models in the table above. The code then prints each response so we can compare and contrast how each model responds to the same prompt.

```
prompts = 'tagline for ice cream shop'
response1 = openai.Completion.create(model="text-davinci-002", prompt=prompts)
response2 = openai.Completion.create(model="text-curie-001", prompt=prompts)
response3 = openai.Completion.create(model="text-babbage-001", prompt=prompts)
response4 = openai.Completion.create(model="text-ada-001", prompt=prompts)

print('davinci-002')
print('-----')
print(response1['choices'][0]['text']).strip(), '\n')

print('curie-001')
print('-----')
print(response2['choices'][0]['text']).strip(), '\n')

print('babbage-001')
print('-----')
print(response3['choices'][0]['text']).strip(), '\n')

print('ada-001')
print('-----')
print(response4['choices'][0]['text']).strip(), '\n')
```

TRY IT!

```
sk-K5M7DDcZgijZ0V9IVSHjT3B1bkFJMqbkLgMiqpzSDsWhxtru
Traceback (most recent call last):
  File "temp.py", line 8, in <module>
    response = openai.Completion.create(model="text-davinci-002", prompt=prompts)
  File "/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_resources/completion.py", line 25, in create
    return super().create(*args, **kwargs)
  File "/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_resources/abstract/engine_api_resource.py", line 115, in create
    response, _, api_key = requestor.request(
  File "/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_requestor.py", line 181, in request
    resp, got_stream =
  self._interpret_response(result, stream)
  File "/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_requestor.py", line 396, in _interpret_response
    self._interpret_response_line(
  File "/home/codio/anaconda3/lib/python3.8/site-packages/openai/api_requestor.py", line 429, in _interpret_response_line
    raise self.handle_error_response(
openai.error.RateLimitError: You exceeded your current quota, please check your plan and billing details.
```

Whereas responses between `curie` and `davinci` are pretty close, we can see a clear difference between the `ada` model and the `davinci` model. Use this tool to generate more comparisons if you want to experiment further.

Fill in the blank with the right model.

- `davinci` is the most capable GPT-3 model.
- `ada` is the fastest and lowest cost model
- `davinci` uses the most recent set of data for training

The `davinci` model is the most capable and has the most recent training data. The `ada` model is the fastest and lowest cost model of the four.

Check it!

Next

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https://codio.com/sandipen-dey/introduction-to-gpt-3/guides/7